IAEA TRS-398 · REFERENCE DOSIMETRY FOR EXTERNAL BEAMS

From chamber reading to absorbed dose

How a calibrated ionization chamber reading becomes absorbed dose to water, for MV photon and electron beams.

$$D_{w,Q} = M_Q \times N_{D,w,Q_o} \times k_{Q,Q_o}$$

$D_{w,Q}$ — Absorbed dose to water
in beam quality Q, at z_{ref} (Gy/MU)

M_Q — Corrected reading
M_1 × all correction factors (nC/MU)

N_{D,w,Q_o} — Calibration coefficient
from the standards lab, in ^{60}Co (Gy/nC)

k_{Q,Q_o} — Beam quality factor
converts the ^{60}Co calibration to your beam

The workflow in five steps

STEP 1 — Measure beam quality
Photons: $TPR_{20,10}$
Electrons: R_{50} (g/cm²)

STEP 2 — Set up in water
Chamber at z_{ref}, 10×10 cm² field, SSD or SAD of 100 cm

STEP 3 — Take readings
M_1, M_+ / M_-, M_2 at V_2, plus air T and P

STEP 4 — Apply corrections
$k_{TP} \cdot k_{elec} \cdot k_{pol} \cdot k_s$ turn M_1 into M_Q

STEP 5 — Calculate dose
× $N_{D,w}$ × k_Q gives Gy/MU; × MU/min gives the dose rate

The corrections that turn M_1 into M_Q

Each one is close to 1. A large deviation usually means a measurement problem.

k_{TP} — Air density

The chamber is open to the air. Scales the reading back to the reference temperature and pressure on the certificate.

$$\frac{273.2 + T}{273.2 + T_0} \times \frac{P_0}{P}$$

Typical **0.97 – 1.03**

k_{pol} — Polarity

The reading changes slightly when the chamber voltage is reversed. Averages both polarities.

$$\frac{|M_+| + |M_-|}{2\,M_1}$$

Typical **0.998 – 1.002**

k_s — Ion recombination

Some ions recombine before they are collected. Uses the two-voltage method (V_1/V_2 between 2 and 5).

$$a_0 + a_1 \frac{M_1}{M_2} + a_2 \left(\frac{M_1}{M_2}\right)^2$$

Typical **1.000 – 1.010**

k_{elec} — Electrometer

Calibration factor of the electrometer when it was calibrated separately from the chamber.

from certificate

If calibrated together **1.000**

$$M_Q = M_1 \times k_{TP} \times k_{elec} \times k_{pol} \times k_s \quad \cdot \text{ readings normalised per MU}$$

Photon vs electron beams

	Photons (MV)	Electrons (MeV)
Beam quality index	$TPR_{20,10} = 1.2661 \cdot PDD_{20,10} - 0.0595$	R_{50} (g/cm²) $= 1.029 \cdot I_{50} - 0.06$
Reference depth z_{ref}	10 g/cm²	$0.6 \cdot R_{50} - 0.1$ g/cm²
Field & distance	10×10 cm² at SSD or SAD 100 cm	≥ 10×10 cm² (≥ 20×20 if $R_{50} > 7$) at SSD 100 cm
Chamber type	Cylindrical (e.g. NE 2571, PTW 30001)	Plane-parallel (e.g. NACP-02, Roos); required if $R_{50} < 4$
k_{Q,Q_o} range	≈ 0.95 – 1.00	≈ 0.88 – 0.94

Worked example · 6 MV photons

NE 2571 chamber, $N_{D,w}$ = 0.0452 Gy/nC, $TPR_{20,10}$ = 0.670, T = 21.5 °C, P = 100.9 kPa, V_1/V_2 = 300/100 V. Illustrative numbers only.

1	M_1 main reading	`0.1480 nC/MU`
2	k_{TP} 21.5 °C, 100.9 kPa	`× 1.0093`
3	k_{pol} $M_\pm$ = 0.1481 / 0.1479	`× 1.0000`
4	k_s M_2 = 0.1474 at 100 V	`× 1.0040`
5	M_Q corrected reading	`= 0.1500 nC/MU`
6	× $N_{D,w}$ × k_Q 0.0452 × 0.998	`× 0.04511`
✓	$D_{w,Q}$ at 10 g/cm²	`0.00677 Gy/MU`

At 600 MU/min this is a dose rate of **4.06 Gy/min**.

Common pitfalls

Let the chamber settle
Allow thermal equilibrium in the water and pre-irradiate the chamber before recording readings.

Watch the units
Pressure in kPa, temperature in °C, readings per MU. Mixing hPa or nC per 100 MU shifts the result.

Stay inside the tables
k_Q and the k_s coefficients are only valid within their tabulated ranges. Don't extrapolate.